

In our context, the above proposition tells us *precisely* how much we can deduce from examining *all* possible relations generated by looking at just a single element $x \in R$, the polynomial g_n being the *minimal* relation from which all others follow. Even though the proof uses some mild ring theory, it sets up equational proofs by generating relations using all $h \in X\mathbb{F}_2[X]$ of degree smaller than n and finding g_n by performing Euclid's algorithm. Any deductions beyond this must be made using at least two variables, as is exemplified by our brief sketch earlier of the proof for Boolean rings.

Further, for any positive integer m , there is a finite field $\mathbb{F}_{2^m}$, which is a $J(2^m)$ -ring. It is easy to see that $\mathbb{F}_{2^m}$ is a $J(n)$ -ring if and only if $m \in S_n$. Any attempt at a reduction to any smaller exponent finds an obvious obstruction in the existence of rings in which this reduction step would fail to hold. One way to interpret the proposition above is that, for any obstruction ring from the set $\{\mathbb{F}_{2^m} \mid m \in S_n\}$, we get a factor contributing to g_n and together these exactly form the minimal relation. In this sense, the obstructions coming from finite fields are really the only ones and our reduction step is optimal.

3 Densities for equational proofs

In order to compute densities of even numbers satisfying the condition that S_n is equal to a given set, we require the following elementary lemma.

Lemma 2. *Let $m, n \in \mathbb{N}$ and denote $d = \gcd(m, n)$. Then*

$$\gcd(2^m - 1, 2^n - 1) = 2^d - 1.$$

Proof. This is a restatement of the fact that $\mathbb{F}_{2^m}^\times \cap \mathbb{F}_{2^n}^\times = \mathbb{F}_{2^d}^\times$. □

Of particular interest to us is the case in which among all the finite fields which are $J(n)$ -rings, there is one that contains all others. This allows us to explicitly reduce the problem to a smaller exponent.

Theorem 3. *Let n be an even positive integer with the property that there is some $m \in \mathbb{N}$ such that all finite $J(n)$ -fields are $\mathbb{F}_{2^m}$ and its subfields. Then any $J(n)$ -ring is also a $J(2^m)$ -ring. The density of even n for which any $J(n)$ -ring is necessarily Boolean is given by*

$$\alpha := \prod_{p \text{ prime}} \frac{2^p - 2}{2^p - 1} \approx 0.54830.$$

Proof. For the first claim, we are to show that $g_n = X^{2^m} - X$. Indeed, by Proposition 1, this is equivalent to showing that $X^{2^k} - X \mid X^{2^m} - X$ for any $k \in S_n$. But by assumption, $\mathbb{F}_{2^k} \subset \mathbb{F}_{2^m}$, proving the claim. From this it follows that for any even positive integer n , any $J(n)$ -ring must be Boolean if and only if $m = 1$ in the previous claim. This happens if and only if $n - 1$ is divisible by no number of the form $2^k - 1$ for $k \geq 2$. Because for any prime $p \mid k$ we have $2^p - 1 \mid 2^k - 1$ by Lemma 2, it suffices for n to not be divisible by any number of the form $2^p - 1$. By the same lemma, all these numbers are pairwise coprime. Since the density of the numbers divisible by $2^p - 1$ is its reciprocal, an expansion using *inclusion-exclusion* yields that the set of numbers not divisible by any of the numbers $\{2^p - 1 \mid p \text{ is prime}\}$ has density

$$\prod_{p \text{ prime}} \left(1 - \frac{1}{2^p - 1}\right),$$

which precisely equals α . Because all the numbers $2^p - 1$ are odd, it does not matter for the density if we restrict to the odd numbers $n - 1$. □

Recall that Morita in [Mor78] found a purely equational proof for the commutativity of all $J(4)$ -rings and all $J(8)$ -rings, of which the latter is much more difficult than the former. If we assume these results, we may drastically increase the density of even n for which explicit equational proofs are known.

Theorem 4. *With the results from [Mor78], Theorem 3 yields equational proofs for the commutativity of all $J(n)$ -rings for a density of even integers n equal to*

$$\frac{571}{365}\alpha \approx 0.85775.$$

Proof. Using Theorem 3 for $m \leq 5$ in combination with [Mor78], we can find equational proofs whenever $S_n \subset \{1, 2\}$ or $S_n \subset \{1, 3\}$. For the first case, we replace the condition of not being divisible by $2^{2^1} - 1 = 3$ by not being divisible by $2^{2^2} - 1 = 15$, yielding a density of $7\alpha/5$. For the second case case, we replace the condition of not being divisible by $2^{3^1} - 1 = 7$ by not being divisible by $2^{3^2} - 1 = 511$, yielding a density of $85\alpha/73$. Subtracting the doubly counted numbers with $S_n = \{1\}$ with density α , the result follows. $\square$

In other words, for about 85.775% of positive even exponents, we can find an explicit equational proof for the commutativity of all $J(n)$ -rings. One may try to explore equational proofs for more families of conditions sufficient for commutativity, some of which are listed in Pinter-Lucke's paper [PL07]. One may further attempt to find equational proofs for odd exponents n , but the loss of the quick observation that $\text{char}(R) = 2$ significantly complicates matters here. The author would welcome any ideas to expand upon these naive attempts further.

4 References

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